

CSE546 - Applied Cryptography

Assignment 1

Due Date - 19th January 2026

Total Marks - 100

Answer any 5 questions.
All question carries 20 marks each.

Question 1. An intelligence agency intercepted an encrypted message. Through side-channel analysis, they discovered the following about the encryption scheme:

The plaintext (spaces removed) is written row by row into a grid with 6 columns. If the plaintext length is not a multiple of 6, pad with the letter Z at the end. The columns are then read out in a secret permutation order π of 1, 2, 3, 4, 5, 6. The ciphertext is the concatenation of columns read top-to-bottom in the order $\pi(1), \pi(2), \dots, \pi(6)$.

The agency also recovered partial information about π :

- $\pi(1) = 4$ (the first column read is column 4)
- $\pi(3) = 1$ (the third column read is column 1)
- $\pi(6) = 2$ (the last column read is column 2)

They know the plaintext is: **ATTACK POSTPONED UNTIL FURTHER ODERS**

And the ciphertext they intercepted is: **ATUOZCPNRDZAPNIHRKOTTEZTSDFRZTOE-LES**

1. Determine the complete permutation π . Show your working clearly.
2. The agency wants to send a decoy message using the same key. Encrypt the following plaintext using the permutation π you found: **ABORT MISSION IMMEDIATELY**
3. Suppose an adversary knows only that a 6-column transposition was used (but not π). How many possible permutations must they try in the worst case? If each decryption attempt takes 1 microsecond, how long would a brute-force attack take?
4. Why does this cipher provide no frequency-based cryptanalysis resistance, yet still offer security against simple pattern matching? Contrast with Vigenère cipher.

Question 2. The Hill cipher works over the alphabet $\{A = 0, B = 1, \dots, Z = 25\}$.

A sender uses a 3×3 Hill cipher key matrix K to encrypt messages. It is known that the following plaintext–ciphertext pairs were generated using the same key matrix K :

- Plaintext: ACT $\rightarrow$ Ciphertext: POH
- Plaintext: NOW $\rightarrow$ Ciphertext: YQF

Question:

Find the key matrix K . If multiple solutions are possible, find any one valid key matrix.

Note: All calculations are to be performed modulo 26. You may assume that the required inverse matrices exist.

Question 3. Answer the following questions with detailed explanations. Support your arguments with appropriate analogies where helpful to illustrate the underlying concepts.

1. Prove that any encryption scheme which provides perfect secrecy must use a key space whose size is at least as large as the message space. Your argument must be information-theoretic and must not rely on any computational hardness assumptions. Use an appropriate analogy to explain why this constraint is fundamental and unavoidable.
2. Argue formally why no symmetric-key encryption scheme can achieve perfect secrecy if the same secret key is reused for encrypting multiple messages, regardless of how the encryption algorithm is designed. Explain this limitation using a concrete analogy that demonstrates why key reuse fundamentally compromises perfect secrecy.

Question 4. Case Study: Security Analysis of a Corporate Communication System

TechSecure Inc., a financial services company, has implemented a custom encryption system for securing inter-departmental communications. The security team made the following design choices:

System Specifications:

- The system uses a 2×2 Hill cipher with a fixed key matrix K to encrypt 2-letter blocks
- All messages use only uppercase letters (A–Z), mapped to $\{0, 1, \dots, 25\}$
- The same key matrix K is used for all communications across the company
- To “enhance security,” the company keeps the encryption algorithm secret
- The key space consists of all 2×2 matrices with determinant coprime to 26

Intercepted Data:

An external auditor performing a security assessment intercepted the following encrypted communications and discovered two plaintext–ciphertext pairs through social engineering:

- Plaintext: HI $\rightarrow$ Ciphertext: JL
- Plaintext: NO $\rightarrow$ Ciphertext: FQ

Your Task:

As the lead security consultant, you must provide a comprehensive security assessment:

1. Cryptanalysis Task:

Using the intercepted plaintext–ciphertext pairs, determine the secret key matrix K used by TechSecure Inc. Show all mathematical steps including:

- Setting up the plaintext and ciphertext matrices
- Computing the required modular inverse
- Finding the key matrix K
- Verifying your answer with both given pairs

2. Key Space Analysis:

The company claims their system is secure because “guessing the right key is computationally infeasible.” Calculate the actual size of the key space (number of valid 2×2 matrices with determinant coprime to 26). Based on your cryptanalysis in part (a), explain why the size of the key space is irrelevant when an attacker has access to known plaintext–ciphertext pairs.

All modular arithmetic calculations should be performed modulo 26.

Question 5. Shannon Theory: Spurious Keys and Perfect Secrecy

Suppose (P, C, K, E, D) is a cryptosystem where the plaintext space P and ciphertext space C satisfy $|P| = |C|$, and keys are chosen equiprobably from the key space K . Let R_L denote the redundancy of the underlying language.

1. Prove that for a ciphertext string of length n , the expected number of spurious keys $\bar{s}_n$ (keys that decrypt the ciphertext into a meaningful but incorrect plaintext) satisfies the following lower bound:

$$\bar{s}_n \geq \frac{|K|}{|P|^{nR_L}} - 1$$

2. Using this result, explain why the One-Time Pad achieves perfect secrecy and why its unicity distance is effectively infinite, preventing an adversary from ever uniquely determining the encryption key.

Question 6.

1. State **Shannon's definition of perfect secrecy** for an encryption scheme.
2. Prove that the **One-Time Pad (OTP)** satisfies Shannon's notion of perfect secrecy.
3. Suppose Alice and Bob communicate using the One-Time Pad. Alice mistakenly uses the *same key* k to encrypt two messages m_1 and m_2 , each of length n , producing ciphertexts c_1 and c_2 . Explain what information Eve gains about the messages.
4. Despite achieving perfect secrecy, explain why the One-Time Pad is **not used in practice**.